CPU → 10^{10} words / sec

MR = 5% ; 10% writes

cache → 1 block = 8 words

BUS width → 2 words

Bandwidth → 10^{12} words / sec

15% blocks are dirty

a) WB + WA

la hit la WB n-ai treabă cu BUS-ul

$$10^{10} \cdot 0{,}05 \cdot (\underset{\text{writes}}{0{,}1} \cdot \text{Write Miss Penalty}^{b}_{WB} + \underset{\text{read miss}}{0{,}9} \cdot \text{Read Miss Penalty}^{b}_{WB})$$

doar la WB+WA $\quad$ Write Miss Penalty$^{b}_{WB}$ = Read Miss Penalty$^{b}_{WB}$

$$= \underbrace{0{,}15 \cdot \frac{8}{2} \text{ Bus Writes}}_{\text{update}} + \underbrace{\frac{8}{2} \text{ Bus reads}}_{\text{allocate}} = 4{,}6 \text{ bus accesses}$$

freq. acese la bus cauzate de miss-uri

$$\Rightarrow 10^{10} \cdot 0{,}05 \cdot 4{,}6 = 0{,}23 \cdot 10^{10}$$

$$\%\ \text{Bus used}_{WB} = \frac{0{,}23 \cdot 10^{10}}{10^{12}} = \frac{0{,}23}{100} = 0.23\%$$

b) WT + WA

$$10^{10} \times \overset{MR}{0{,}05} \cdot (0{,}1 \cdot \text{Write Miss Penalty}_{WT} + 0{,}9 \cdot \text{Read Miss Penalty}_{WT}) +$$

la WT ai miss și la hit (scrie și în cache și în MM)

$$+ 10^{10} \cdot \underset{\text{freq hits}}{0{,}95} \cdot \underset{\text{writes}}{0{,}1} \cdot \text{Write Hit Penalty}_{WT}$$

$$\text{Write Miss Penalty}^{b}_{WT} = \underbrace{\frac{8}{2} \text{ Bus Reads}}_{\text{Allocate}} + \underset{MM}{1 \text{ Bus Write}} = 5 \text{ bus accesses}$$

$$\text{Read Miss Penalty}^{b}_{WT} = \underbrace{\frac{8}{2} \text{ Bus Reads}}_{\text{allocate}} = 4 \text{ bus accesses}$$

Write Hit Penalty$^{B}_{WT}$ = 1 bus write

$$\Rightarrow 10^{10} \cdot 0{,}05 \cdot (0{,}1 \cdot 5 + 0{,}9 \cdot 4) + 10^{10} \cdot 0{,}95 \cdot 0{,}1 \cdot 1 =$$

$$= 10^{10} \cdot 0{,}205 + 10^{10} \cdot 0{,}095 = 0{,}3 \cdot 10^{10}$$

% bus used$_{WT}$ = 0,3%